tree WuT'. After $Explore(E.p, Sleep', WuT')$ has returned, the sleep set $sleep(E)$ is extended with p, and all sequences beginning with p are removed from $wut(E)$.

7.2 Correctness

Let us now prove the correctness of the optimal-DPOR algorithm. Throughout, we assume a particular completed execution of optimal-DPOR. This execution consists of a number of terminated calls to $Explore(E, Sleep, WuT)$ for some values of the parameters E, $Sleep$, and WuT. Let $\mathcal{E}$ denote the set of execution sequences E that have been explored in some call $Explore(E, \cdot, \cdot)$. Define the ordering $\propto$ on $\mathcal{E}$ by letting $E \propto E'$ if $Explore(E, \cdot, \cdot)$ returned before $Explore(E', \cdot, \cdot)$. Intuitively, if one were to draw an ordered tree that shows how the exploration has proceeded, then $\mathcal{E}$ would be the set of nodes in the tree, and $\propto$ would be the post-order between nodes in that tree.

For an arbitrary execution sequence $E \in \mathcal{E}$ let:

- $final_sleep(E)$ denote the value of $sleep(E)$ at the point when $Explore(E, Sleep, WuT)$ returns.
- $done(E)$ denote $final_sleep(E) \setminus Sleep$, i.e., the set of processes that are explored from $s_{[E]}$.

We begin by establishing some useful invariants.

Lemma 7.1. *Whenever Algorithm 2 is inside a call $Explore(E, Sleep, WuT)$, then the following two invariants hold:*

(1) $wut(E)$ is a wakeup tree after $\langle E, sleep(E)\rangle$,
(2) if WuT is empty, then $Sleep$ is empty.

Proof. We establish these two invariants jointly, by induction over the steps of the algorithm. The properties hold at the beginning of the initial call, since $sleep(E)$ and $wut(E)$ are both empty. We verify that each property is preserved by the steps of the algorithm.

(1) We need to consider the following cases.
- Whenever $wut(E)$ is updated at line 7, it follows by Property 1 of the operation $insert_{[E]}(v, wut(E))$ that $wut(E)$ remains a wakeup tree after $\langle E, sleep(E)\rangle$.
- Consider a new call to $Explore(E.p, Sleep', WuT')$ at line 19, which occurs inside a call $Explore(E, Sleep, WuT)$, and where $Sleep'$ and WuT' are obtained from the current values of $sleep(E)$ and $wut(E)$ by $Sleep' = \{q \in sleep(E) \mid E \vdash p \diamond q\}$ and $WuT' = subtree(wut(E), p)$, at lines 17 and 18. Since the parameters $Sleep'$ and WuT' will be used to initialize $sleep(E.p)$ (at line 14) and $wut(E.p)$ (at line 10), we must check that WuT' is a wakeup tree after $\langle E.p, Sleep'\rangle$. By the inductive hypothesis, $wut(E)$ is a wakeup tree after $\langle E, sleep(E)\rangle$. We establish that WuT' is a wakeup tree after $\langle E.p, Sleep'\rangle$ by checking the two properties of Definition 6.1. Let $WuT' = \langle B', \prec'\rangle$ and let $wut(E) = \langle B, \prec\rangle$.
  (a) Let w be a leaf of B'; then $p.w$ is a leaf of B. Let $q \in Sleep'$. By the update at line 17, we have $q \in sleep(E)$ and $E \vdash p \diamond q$. By the inductive hypothesis $q \notin WI_{[E]}(p.w)$, which using $E \vdash p \diamond q$ implies $q \notin WI_{[E.p]}(w)$. Hence $WI_{[E.p]}(w) \cap Sleep' = \emptyset$ whenever w is a leaf of B'.
  (b) Let $u.q$ and $u.w$ be nodes in B' with $u.q \prec' u.w$, and $u.w$ is a leaf. Then $p.u.q$ and $p.u.w$ are nodes in B with $p.u.q \prec p.u.w$, and $p.u.w$ a leaf. By the inductive hypothesis, it follows that $q \notin WI_{[E.p.u]}(w)$, which is precisely what we must establish for the inductive step.

  If $wut(E.p)$ is initialized at line 13 instead of at line 10, then this initialization is performed in a call of form $Explore(E.p, Sleep', WuT')$ where WuT' is the empty wakeup tree. By Invariant 2 of the inductive hypothesis, $Sleep'$ is empty. Since $\langle\{\langle\rangle, p\}, \{(p, \langle\rangle)\}\rangle$ is trivially a wakeup tree after $\langle E.p, \emptyset\rangle$, the property follows.